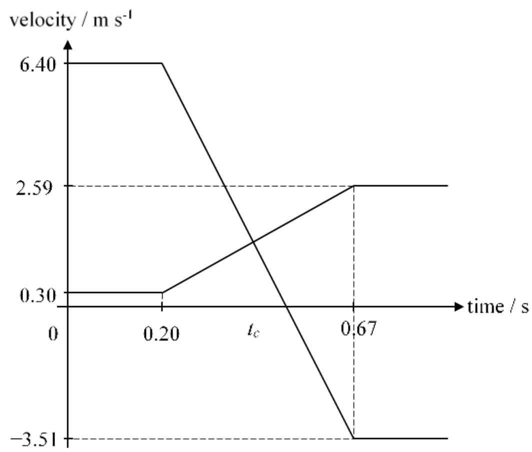
2	
(a)	The total final momentum of a system after a collision is equal to the total initial momentum of a system before the collision
	provided that the net external force acting on the system is zero.
(b)(i)	$m_A u_A + m_B u_B = m_A v_A + m_B v_B$ $(1.2)(6.40) + (5.2)(0.30) = 1.2 v_A + 5.2 v_B$
	$u_A - u_B = v_B - v_A$ $v_A = v_B - 6.10$ or $v_B = v_A + 6.10$
	$1.2 v_A + 5.2(v_A + 6.10) = 9.24$ or $1.2(v_B - 6.10) + 5.2 v_B = 9.24$ $v_A = -3.5125 \text{ m s}^{-1}$ $v_B = 2.5875 \text{ m s}^{-1}$
	$= 2.59 \text{ m s}^{-1}$ (shown)
(b)(ii)	Considering B
	$\Delta p = (5.2)(2.59 - 0.3)$ $= 11.895$
	$F = \dfrac{\Delta p}{t}$ $= \dfrac{11.895}{0.47} = 25.31 \text{ N}$
	Using Newton's 3rd law, force A exerts on B is equal in magnitude but opposite in direction to the force B exerts on A, therefore the force B on A will be -25.31N
b)(iii)	
	Correct label for u_A, v_A and straight line before t and after t + 0.47
	Straight line with negative slope between t and t + 0.47

(b)(iv)	t_c at intersection between the 2 lines
(b)(v)	Kinetic energy of the system is conserved before and after elastic collisions
	At t_c, some of the kinetic energy is converted to elastic potential energy and stored in the spring.

3	